

On a Conjecture on the Values of ψ -Correlators

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1 Overview

The purpose of this paper is to outline some results on the numerical values of ψ -correlators on the cohomology of the moduli space of pointed stable curves $\overline{\mathcal{M}}_{g,n}$. The main conjecture aimed to be proven was

Conjecture 1 (Liu, Xu [2]). *If $\sum d_i = 3g - 3 + n$ and $d_1 < d_2$, then*

$$\langle \tau_{d_1} \tau_{d_2} \cdots \tau_{d_n} \rangle_g \leq \langle \tau_{d_1+1} \tau_{d_2-1} \cdots \tau_{d_n} \rangle_g.$$

In other words, as the indices of the correlators become more evenly distributed, the value becomes larger. Some partial results towards this conjecture were obtained including a proof of the statement for $n = 2$, and a proof of an analogous statement for a related generating function for $n = 3$, to be explained below. The outline of this paper is to provide some background, although it will be the minimum needed to explain notation, and then state and sketch proofs for the results. The results have not been proven completely, but they have been checked in enough cases that their validity should hold in general. Places where statements have not been proven completely will be outlined, and either evidence towards the claim, or outlines for what must be done to complete a proof will be provided.

2 Background

The moduli space $\overline{\mathcal{M}}_{g,n}$ is a parameter space for complex structures on genus g curves with n marked points. Each point on $\overline{\mathcal{M}}_{g,n}$ corresponds to a complex curve C of arithmetic genus g which is allowed to have singularities, provided they are nodal, and an n -tuple of marked points $p_1, \dots, p_n \in C$ with the additional stability condition that in the normalization of a singular curve, every genus 0 component must have at least 3 special points (marked points or pre-image of a node) and every genus 1 component must have at least 1 special point. In particular $2g - 2 > 0$.

For every marked point p_i , there is an associated line bundle on $\overline{\mathcal{M}}_{g,n}$ where the fiber over $[C, \{p_1, \dots, p_n\}]$ representing C and the n -tuple $p_1, \dots, p_n$ is the cotangent vector space of C at the point p_i . The first Chern class of this line bundle is denoted ψ_i , and is an element of the first cohomology of $\overline{\mathcal{M}}_{g,n}$. The dimension of $\overline{\mathcal{M}}_{g,n}$ is $3g - 3 + n$, so we can take the intersection numbers of these cohomology classes obtaining the correlators

$$\langle \tau_{d_1} \dots \tau_{d_n} \rangle_g = \int_{\overline{\mathcal{M}}_{g,n}} \psi_1^{d_1} \dots \psi_n^{d_n}$$

when $\sum d_i = 3g - 3 + n$. ψ -classes generate a large part of the cohomology of $\overline{\mathcal{M}}_{g,n}$, so understanding these intersection numbers is desirable to understand the geometry of the moduli space. The computation of these numbers is possible through a recursive method by considering an infinite variable generating function encoding all correlators. It is Witten's conjecture that this generating function satisfies the so-called KdV hierarchy, and this was proved first by Kontsevich, and later by others. Since we will not directly use this formulation of the recursion, we omit it, but instead use an alternative method for calculation found by Liu and Xu.

We denote the n -point generating function as

$$F(x_1, \dots, x_n) = \sum \langle \tau_{d_1} \dots \tau_{d_n} \rangle x_1^{d_1} x_2^{d_2} \dots x_n^{d_n}$$

and the “normalized” generating function [cite] as

$$G(x_1, \dots, x_n) = \exp \left(-\frac{x^3 + y^3 + z^3}{24} \right) F(x_1, \dots, x_n).$$

Liu and Xu [3] provide explicit formulas for F and G for $n = 2, 3$, and give a recursive method for computing them for higher n . We will mainly use the generating function for $n = 3$, and relabel the coefficients as x, y, z instead of x_1, x_2, x_3 . For an arbitrary function H and monomial X , we denote $H[X]$ for the coefficient of X in H . In this notation, the conjecture can be rewritten as $F[x_1^{a+1}x_2^{b-1}Y] \geq F[x_1^a x_2^b Y]$ whenever $a < b$ where Y is any fixed monomial in $x_3, \dots, x_n$ such that $\deg(Y) = 3g - 3 + n - (a + b)$. Going forward, we use n to denote an index of the ψ -correlators rather than the number of points. In the case of 3 correlators, $Y = z^n$ for some n .

2.1 Results

Zograf [1] found an explicit formula for 2-correlators. By manipulating this expression and using some numerical estimates, we prove Liu and Xu's claim for 2-correlators. We also show that $G[x^{a+1}y^{b-1}z^n] \geq G[x^a y^b z^n]$ whenever $a < b$ except when $g < n$ or $g = n = 1$. This is done by finding an alternative expression for $G(x, y, z)$ which allows for the comparison of binomial coefficients. While this is not equivalent to the conjecture, the method used to prove this gives that the inequality holds exactly when $a < b$, so it seems suggestive that this can be used to prove the conjecture for 3-correlators. It is desirable to know a geometric relationship between F and G which could allow us to prove the statement for F after knowing it for G . The exponential factor involving x_i^3 and $1/24 = \langle \tau_1^3 \rangle$ is highly suggestive of some genus 1 behavior.

3 First Result: 2-Correlators

Here, we prove the conjecture of Liu and Xu in the simplest case, that of two correlators:

Theorem 3.1. *For $n > m$, $\langle \tau_n \tau_m \rangle \leq \langle \tau_{n-1} \tau_{m+1} \rangle$.*

Proof. Zograf [1] showed

$$24^g g! \langle \tau_k \tau_{3g-k-1} \rangle = \frac{(6g-1)!!}{(2k+1)!!(6g-1-2k)!!} \left(\frac{6g-3}{6g-1} + \sum_{i=0}^{k-1} b_{g,i} \right)$$

where b_i are defined as

$$b_{g,k} = \frac{(6g-3-2k)!!(2k+1)!!}{(6g-1)!!} \begin{cases} \frac{(g-1)!}{j!(g-j)!} (g-2j) & \text{if } k = 3j-1 \\ -2 \frac{j!(g-j-1)!}{j!(g-j-1)!} & \text{if } k = 3j \\ 2 \frac{(g-1)!}{j!(g-j-1)!} & \text{if } k = 3j+1 \end{cases}$$

which implies

$$24^g g! \langle \tau_k \tau_{3g-k-1} \rangle = 24^g g! \frac{6g+1-2k}{2k+1} \langle \tau_{k-1} \tau_{3g-k} \rangle + \frac{(6g-1)!!}{(2k+1)!!(6g-1-2k)!!} b_{g,k-1}$$

which is obtained since the expression for $24^g g! \frac{6g+1-2k}{2k+1} \langle \tau_{k-1} \tau_{3g-k} \rangle$ accounts for all terms in the expansion for $\langle \tau_k \tau_{3g-k-1} \rangle$ except for the last term. If $3g-k > k-1$, then $\langle \tau_k \tau_{3g-k-1} \rangle$ is more equilibrated than $\langle \tau_{k-1} \tau_{3g-k} \rangle$, so we want to show $\langle \tau_k \tau_{3g-k-1} \rangle > \langle \tau_{k-1} \tau_{3g-k} \rangle$. For this, consider

$$24^g g! (\langle \tau_k \tau_{3g-k-1} \rangle - \langle \tau_{k-1} \tau_{3g-k} \rangle) = 24^g g! \frac{6g-4k}{2k+1} \langle \tau_{k-1} \tau_{3g-k} \rangle + \frac{(6g-1)!!}{(2k+1)!!(6g-1-2k)!!} b_{g,k-1}.$$

It suffices to show the right side of this equality is nonnegative for $3g-k > k-1$, since if $3g-k \leq k-1$, then the case has already been covered since if we let $k' = 3g-k-1$, $3g-k' > k'-1$ and now, it is $\langle \tau_{k'} \tau_{3g-k'-1} \rangle$ which is more equilibrated, but $2k \geq 3g+1$ so $k' \leq k$. Now in this range, if $k \not\equiv 1 \pmod{3}$, $6g \geq 4k$ and Zograf showed that $b_{g,k-1}$ is positive. It is known that all correlators are positive. Then, both terms on the right side are nonnegative and the claim holds.

Now suppose $k \equiv 1 \pmod{3}$. In what follows, $k = 3j+1$. The expression for $b_{g,k-1}$ is

$$b_{g,k-1} = -\frac{(6g-3-2(k-1))!!(2k+1)!!}{(6g-1)!!} \frac{2(g-1)!}{j!(g-1-j)!}.$$

Also from Zograf, we have

$$24^g g! \langle \tau_{k-1} \tau_{3g-k} \rangle = \frac{(6g-1)!!}{(2k-1)!!(6g-2k+1)!!} a_{g,k-1} \geq \frac{(6g-1)!!}{(2k-1)!!(6g-2k+1)!!} \frac{6g-3}{6g-1}.$$

Furthermore, in the range we care about, since g is an integer, we can strengthen the inequality on g to $g \geq \frac{2k+1}{3}$ (since $2k+1$ is the smallest multiple of 3 at $\geq 2k-1$) implying $6g-4k \geq 2$. Thus, to prove our claim, it suffices to show the middle inequality in

$$\begin{aligned} & \left| \frac{(6g-1)!!}{(2k+1)!!(6g-1-2k)!!} b_{g,k-1} \right| = \frac{2(g-1)!}{j!(g-1-j)!} \\ & \leq \frac{2}{2k+1} \frac{(6g-1)!!}{(2k-1)!!(6g-2k+1)!!} \frac{6g-3}{6g-1} \leq \frac{2}{2k+1} 24^g g! \langle \tau_{k-1} \tau_{3g-k} \rangle \end{aligned}$$

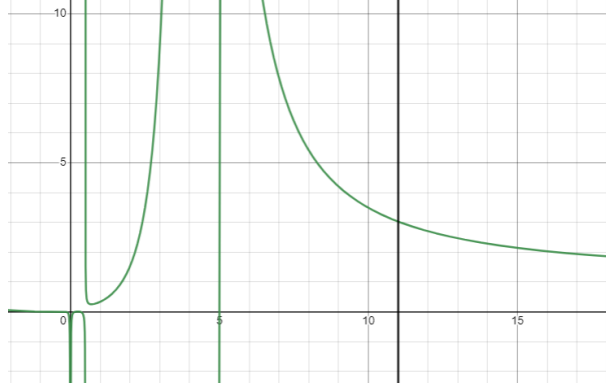


Figure 1: A plot of $F(g, j)$ for $j = 5$ (green) and the line $g = 2j + 1$ (black)

After some cancellation and replacing double factorials with factorials, we obtain that the desired inequality is equivalent to

$$\begin{aligned} \frac{(g-1)!}{j!(g-1-j)!} &\leq \frac{(6g-1)!!}{(2k+1)!!(6g-2k+1)!!} \frac{6g-3}{6g-1} = \frac{(6g-1)!!}{(6j+1)!!(6g-6j-1)!!} \frac{6g-3}{6g-1} \\ \frac{(6j+2)!}{2^{3j+1}(3j+1)!} \frac{(g-1)!}{j!(g-1-j)!} &\leq \frac{(6g)!}{2^{3g}(3g)!} \frac{2^{3g-3j}(3g-3j)!}{(6g-6j)!} \frac{6g-3}{6g-1} \\ \frac{(6j+2)!}{2(3j+1)!} \frac{(g-1)!}{j!(g-1-j)!} &\leq \frac{(6g)!}{(3g)!} \frac{(3g-3j)!}{(6g-6j)!} \frac{6g-3}{6g-1} \end{aligned}$$

To show the last inequality is true, we first show that with fixed j , if it is true for the extremal case $g = \frac{2k+2}{3} = 2j+1$, then it will be true for all $g \geq 2j+1$. To do this, consider the function

$$F(g, j) = \frac{(6g)!(3g-3j)!(6g-3)(g-1-j)!}{(3g)!(6g-6j)!(6g-1)(g-1)!}$$

obtained by moving all the terms involving a g to the right side. The above inequality is hence equivalent to $\frac{(6j+2)!}{2(3j+1)!j!} \leq F(g, j)$. Now,

$$\begin{aligned} \frac{F(g+1, j)}{F(g, j)} &= \frac{(6g+6)(6g+5)(6g+4)(6g+3)(6g+2)(6g+1)(3g+3-3j)}{(6g+6-6j)(6g+5-6j)(6g+4-6j)(6g+3-6j)(6g+2-6j)} \\ &\quad \times \frac{(3g+2-3j)(3g+1-3j)(6g-2)(6g-1)(g-j)}{(6g+1-6j)(3g+3)(3g+2)(3g+1)(6g-3)(6g)(g)}. \end{aligned}$$

Although a formal proof must be worked out, it can be seen graphically that $\frac{F(g+1, j)}{F(g, j)} > 1$ for $g \geq 2j+1$, as is shown in Figure 1. It is easy to see that $F(g, j)$ approaches 1 as $g \rightarrow \infty$ for fixed j since the terms in the numerator match those in the denominator. For varying values of j , the graph lies above 1 for $g \geq 2j+1$. Taking this for granted, $F(g, j)$ is an increasing function of g in this range. Thus, if $\frac{(6j+2)!}{2(3j+1)!j!} \leq F(2j+1, j)$ then $\frac{(6j+2)!}{2(3j+1)!j!} \leq F(g, j)$ for all $g \geq 2j+1$.

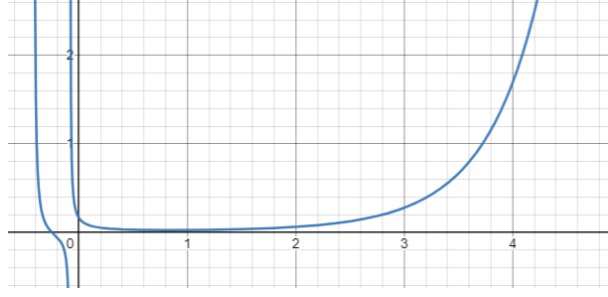


Figure 2: The function $\frac{2 \cdot 4^{2j+2} (12j+3)}{(12j+7)(12j+5)(6j+6)(6j+5)(6j+4)}$

Now, to show the inequality for $g = 2j + 1$, we substitute for g and rearrange

$$\begin{aligned} \frac{(6j+2)!}{2(3j+1)!} \frac{((2j+1)-1)!}{j!((2j+1)-j-1)!} &\leq \frac{(6(2j+1))!}{(3(2j+1))!} \frac{(3(2j+1)-3j)!}{(6(2j+1)-6j)!} \frac{12j+3}{12j+5} \\ \frac{(6j+2)!}{2(3j+1)!} \frac{(2j)!}{j!j!} &\leq \frac{(12j+6)!}{(6j+3)!} \frac{(3j+1)!}{(6j+3)!} \frac{12j+3}{(12j+5)(6j+6)(6j+5)(6j+4)} \\ \frac{1}{2} \binom{6j+2}{3j+1} \binom{2j}{j} &\leq \binom{12j+6}{6j+3} \frac{12j+3}{(12j+5)(6j+6)(6j+5)(6j+4)} \end{aligned}$$

Using the inequalities $\frac{4^n}{2^{n+1}} \leq \binom{2n}{n} \leq 4^n$, we can upper bound the left hand side and lower bound the right hand side to show it suffices to prove

$$\frac{4^{4j+1}}{2} \leq \frac{4^{6j+3}}{12j+7} \frac{(12j+3)}{(12j+5)(6j+6)(6j+5)(6j+4)}$$

This inequality holds for $j \geq 4$ as can be seen in Figure 2 since the graph monotonically increases for $j \geq 4$ (the inequality is equivalent to the function being at least 1). so the claim is true for $g \geq 2(2) + 1 = 9$. For $g < 9$, the values can be checked explicitly using admcycles [4] and the theorem holds. $\square$

4 Second Result: Analog for $G(x, y, z)$

This section sketches an outline of the following

Theorem 4.1. $G[x^{a+1}y^{b-1}z^n] \geq G[x^a y^b z^n]$ when $g \geq n$ except for $g = n = 1$ where $g = \frac{a+b+n}{3}$

Liu and Xu [3] provide an explicit formula for $G(x, y, z)$, originally due to Zagier, as follows:

$$G(x, y, z) = \sum_{r,s \geq 0} \frac{r! S_r(x, y, z)}{4^r (2r+1)!! \cdot 2} \cdot \frac{\Delta(x, y, z)^s}{8^s (r+s+1)!}$$

where

$$P_r(x, y, z) = \frac{(xy)^r (x+y)^{r+1} + (yz)^r (y+z)^{r+1} + (xz)^r (x+z)^{r+1}}{x+y+z}$$

$$\Delta(x, y, z) = (x + y)(y + z)(x + z).$$

We will write this in an alternate form. First, let $g = r + s$ (this agrees with the above definition, as may be clearer later) and note that the constant coefficient can be equivalently written as

$$\begin{aligned} \frac{r!}{4^r(2r+1)!! \cdot 2 \cdot 8^s(r+s+1)!} &= \frac{(g-s)!}{4^{g-s}(2(g-s)-1)!! \cdot 2^{3s+1}(g+1)!} \\ &= \frac{((g-s)!)^2}{2^{2g+s+1} \cdot 2^{-(g-s)}(2(g-s)+1)!(g-s+1)} = \frac{1}{2^{g+2s+1} \binom{2(g-s)}{g-s} (2(g-s)+1)}. \end{aligned}$$

This expression will be useful later to derive estimates. However, for the time being, the value of the quantity does not affect too much and we denote it $C_{g,s}$. Hence, we have

$$G(x, y, z) = \sum_{g,s \geq 0} C_{g,s} S_r(x, y, z) \Delta(x, y, z)^s.$$

The next goal is to rewrite it as

$$G = \sum_{g,s \geq 0} C_{g,s} \sum_{n \geq 0} \frac{z^n}{n!} G_n(x, y)$$

i.e. isolate the z term. This is done because, as explained above, the conjecture is equivalent to comparing the coefficients for monomials with a fixed exponent of z (for F , but here we prove an analog for G). Note that G_n implicitly depends on s and r (equivalently $g = r + s$), but for notational convenience, we suppress this dependence. G_n can be obtained in a straightforward manner by differentiating with respect to z n times:

$$G_n(x, y) = \left. \frac{d^n}{dz^n} \right|_{z=0} S_r(x, y, z) \Delta(x, y, z)^s$$

These derivatives were carried out in `sagemath`, and the following formula is obtained:

$$G_n(x, y) = (xy(x+y))^{g-n} \left(\sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} D_{n,k} s^{\frac{n-k}{2}} (x+y)^{2(n-k)} (xy)^k + (-1)^n n! (xy)^n \right).$$

where

$$D_{n,k} = \frac{n!}{k!(n-2k-1)!(n-k)}$$

and

$$s^r = s(s-1)\dots(s-(r-1))$$

and again, $g = r + s$. While this formula has not yet been proven, it most certainly should be correct: the values of $D_{n,k}$ up to $n = 10$ were computed numerically from which the general formula was extrapolated and checked to be correct for $n = 11$. The above formula is a bit

opaque, so here are the equations for $n = 1$ through $n = 6$ to get a sense of the pattern:

$$\begin{aligned}
G_1 &= (xy(x+y))^{g-1}((x+y)^2s - xy) \\
G_2 &= (xy(x+y))^{g-2}((x+y)^4s(s-1) + 2(xy)^2) \\
G_3 &= (xy(x+y))^{g-3}((x+y)^6s(s-1)(s-2) + 3xy(x+y)^4s(s-1) - 6(xy)^3) \\
G_4 &= (xy(x+y))^{g-4}((x+y)^8s(s-1)(s-2)(s-3) + 8xy(x+y)^6s(s-1)(s-2) + 24(xy)^4) \\
G_5 &= (xy(x+y))^{g-5}((x+y)^{10}s\ldots(s-4) + 15xy(x+y)^8s\ldots(s-3) + 20(xy)^2(x+y)^6s\ldots(s-2) \\
&\quad - 5!(xy)^5) \\
G_6 &= (xy(x+y))^{g-6}((x+y)^{12}s\ldots(s-5) + 24xy(x+y)^8s\ldots(s-4) + 90(xy)^2(x+y)^6s\ldots(s-3) \\
&\quad + 6!(xy)^6)
\end{aligned}$$

A couple things to note: all the numerical coefficients above are positive except for the $-n!$ when n is odd. Additionally, there is no longer any explicit r dependence, and our equation is solely in terms of g and s .

Now, we want to compare $G[x^{a+1}y^{b-1}z^n]$ and $G[x^ay^bz^n]$. As $C_{g,s}/n! > 0$, to prove our claim, it would suffice to show $G_n[x^{a+1}y^{b-1}] \geq G_n[x^ay^b]$ for all n whenever $a < b$. We split this up into two cases, n even and n odd (because of the additional negative term in the latter case).

Case 1 (n even): As remarked earlier, all the numerical coefficients in the expansion of G_n are positive so we can further reduce to showing that the coefficient of $x^{a+1}y^{b-1}$ coming from each term which looks like $(x+y)^{g-n}(x+y)^{2(n-k)}(xy)^k$ is greater than the coefficient of x^ay^b . Let $H = (x+y)^{g+n-2k}(xy)^{g-n+k}$ and $H' = (x+y)^{g+n-2k}$. Then, we have

$$H[x^ay^b] = H'[x^{a-(g-n+k)}y^{b-(g-n+k)}] = \binom{g+n-2k}{a-(g-n+k)}.$$

Note that $a+b = 3g-n$ as that is the total degree of H . Now, for $H[x^{a+1}y^{b-1}]$ to be greater than $H[x^ay^b]$, it is equivalent to

$$a - (g - n + k) < \frac{g + n - 2k}{2}.$$

This is because the binomial coefficients increase up to the half way point where they start decreasing. But this is equivalent to

$$a < \frac{g+n-2k}{2} + g-n+k = \frac{3g-n}{2} = \frac{a+b}{2}$$

which is exactly the statement $a < b$, the condition we wanted! The simplicity of this argument suggests that it is a viable method of proving the conjecture in general.

Case 2 (n odd): This case is a bit messier since we will have to genuinely add things. Recall that

$$G = \sum_{g,s \geq 0} C_{g,s} \sum_{n \geq 0} \frac{z^n}{n!} G_n(x,y) = \sum_{g,n \geq 0} \frac{z^n}{n!} \sum_{s=0}^g C_{g,s} G_n(x,y)$$

where we have rearranged the sums to analyze terms for g and n fixed, and used that since $g = r+s$ and $r \geq 0$, s can take any value between 0 and g . Now, looking above at the expression

for G_n , when $s = 0$ for example, there are no positive terms, but there is still a $-n!(xy)^n(xy(x+y))^{g-n}$, so we have to sum over all s to make see if the negative contributions persist.

First, we use the formula above for $C_{g,s}$ to find useful bounds. Using again that $\frac{4^n}{2n+1} \leq \binom{2n}{n} \leq 4^n$, we obtain the particularly nice bounds

$$\frac{1}{2^{3g+1}(g+1)!(2(g-s)+1)} \leq C_{g,s} = \frac{1}{2^{g+2s+1}\binom{2(g-s)}{g-s}(2(g-s)+1)} \leq \frac{1}{2^{3g+1}(g+1)!}.$$

Now, let $G_n'' = n!(xy)^n(xy(x+y))^{g-n}$ and $G_n' = G_n - G_n''$ (splitting the positive and negative parts). Using these bounds on

$$\sum_{s=0}^g C_{g,s} G_n(x, y) = \sum_{s=0}^g C_{g,s} G_n'(x, y) - \sum_{s=0}^g C_{g,s} G_n''(x, y),$$

we can lower bound the positive part by

$$\sum_{s=0}^g \frac{1}{2^{3g+1}(g+1)!(2(g-s)+1)} G_n'(x, y)$$

and upper bound the negative part by

$$\frac{g}{2^{3g+1}(g+1)!} G_n''(x, y).$$

Hence, factoring out common terms, it suffices to prove the claim for

$$\left(\sum_{s=0}^g \frac{1}{(2(g-s)+1)} G_n'(x, y) \right) - g G_n''(x, y).$$

We have

$$\begin{aligned} G_n'[x^{a+1}y^{b-1}] - G_n'[x^a y^b] &= \sum_k D_{n,k} s^{n-k} \left(\binom{g+n-2k}{a-(g-n+k)+1} - \binom{g+n-2k}{a-(g-n+k)} \right) \\ &= \sum_k D_{n,k} s^{n-k} \left(\frac{(g+n-2k)!(3g-2a-n-1)}{(a-g+n-k+1)!(2g-a-k)!} \right) \end{aligned}$$

and

$$G_n''[x^{a+1}y^{b-1}] - G_n''[x^a y^b] = n! \left(\binom{g-n}{a-g+1} - \binom{g-n}{a-g} \right) = n! \frac{(g-n)!(3g-2a-n-1)}{(a-g+1)!(2g-a-n)!},$$

where the binomial coefficients are derived in a similar fashion to what was done in Case 1. Putting everything together, we have to show that

$$\sum_{s=0}^g \frac{1}{(2(g-s)+1)} \sum_k D_{n,k} s^{n-k} \left(\frac{(g+n-2k)!(3g-2a-n-1)}{(a-g+n-k+1)!(2g-a-k)!} \right) \geq gn! \frac{(g-n)!(3g-2a-n-1)}{(a-g+1)!(2g-a-n)!}$$

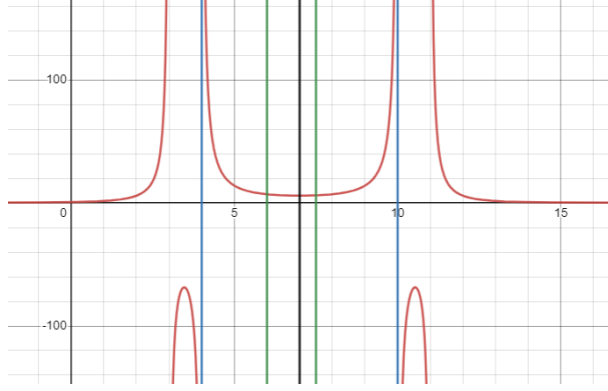


Figure 3: The summand with $g = 6$, $n = 3$, $k = 1$ with lines for the symmetry axis i.e. minimum (black), closest poles at $a = g - 2$ and $a = 2g - (n - 1)$ (blue), and the ends of the range we care about, $a = g$ and $a = \frac{3g-1}{2}$ (green).

in the appropriate range. We only care about when $a < b$ i.e. $a < \frac{3g-n}{2}$, but also note that $G''_n = (xy)^g(x+y)^{g-n}$, so it also suffices to assume $a > g$, otherwise we get no negative terms and the same proof as Case 1 works. The above inequality can be rewritten as

$$\frac{1}{gn!} \sum_{s=0}^g \sum_k \frac{D_{n,k} s^{n-k}}{(2(g-s)+1)} \left(\frac{(g+n-2k)!(a-g+1)!(2g-a-n)!}{(g-n)!(a-g+1+n-k)!(2g-a-k)!} \right) \geq 1.$$

We can remove the a dependence on this inequality as follows (Figure 3): the summand has a symmetry around the value $a = \frac{3g-(n+1)}{2}$. It is easily verified that the denominator vanishes at $a = g - 2$ and $a = 2g - (n - 1)$, and the function is positive in this range so that there is a minimum at $a = \frac{3g-(n+1)}{2}$.

Since we are only considering $g < a < \frac{3g-n}{2}$, which falls into this range, it suffices to replace the summand with this minimum value and prove

$$\frac{1}{gn!} \sum_{s=0}^g \sum_k \frac{D_{n,k} s^{n-k}}{(2(g-s)+1)} \left(\frac{(g+n-2k)!(\frac{g-n-1}{2})!(\frac{g-n-1}{2})!}{(g-n)!(\frac{g+n-1}{2}-k)!(\frac{g+n+1}{2}-k)!} \right) \geq 1.$$

This inequality has not been fully proven, but a sketch is now shown to show why it should be true provided $g \geq n$. In fact, it suffices (or should suffice) to reduce to the summand of $k = 0$ and $s = g$, for which the summand becomes ($D_{n,0} = 1$, as can be seen in the examples above):

$$W(g, n) = \frac{g(g-1) \dots (g-n+1)}{gn!} \left(\frac{(g+n)!}{(g-n)!} \left(\frac{(\frac{g-n+1}{2})!}{(\frac{g+n+1}{2})!} \right)^2 \right).$$

It suffices to show $W(g, n) \geq 1$ for $g \geq n$. Unfortunately, this fails for $g = n = 1$, but numerically seems to hold otherwise. A method to prove this in general would be to first show that $W(n, n) \geq 1$ for all $n \geq 2$ and $W(2, 1) \geq 1$, and then $W(g+1, n) \geq W(g, n)$ for all $g \geq n$. Again, this seems to hold numerically but would take some work to rigorously show.

5 Final Remarks

It was attempted to generalize the results of the first method to more points. As an explicit formula is usually deemed difficult, instead it was attempted to roughly follow the steps leading to the explicit formula for 2-correlators (some recursion) to get some bounds on 3-correlators. Unfortunately, the methods seemed too difficult to get something sensible, leading to the investigation of the generating function itself which led to the second result.

While it does not prove the conjecture, the method of proof in the second result seems much more promising towards something that would lead to a full solution. First, the numerics were much nicer - just by comparing binomial coefficients, we are led to the condition that $a < b$, so in some sense, if the conjecture is true, this way of splitting the generating function seems to “partition the coefficients in just the right way”. In comparison, the bounds used in the first method seemed to not have much structure to them. Additionally, an analogous statement for the two point formula in [3] is even easier leads to the same sort of analysis with binomial coefficients, coming again at the condition $a < b$. Hence it seems likely that such analysis will hold towards higher point correlator generating functions as well. Liu and Xu provide a recursive method for determining $G(x_1, \dots, x_n)$ which can hopefully be manipulated to get similar results.

One thing that must be checked in the above analysis is what happens when $g < n$, since then the $(x + y)^{g-n}$ term appears as “a denominator”, so more careful analysis must be done to see if the theorem holds here as well. As stated before, it is desirable to also understand the geometry of G i.e. give interpretations to the coefficients, and also the indices. One of the indices that was summed over was suggestively called g , this is because it very much acts like the genus for correlators in that the exponents sum to $3g - 3 + 3 = 3g$ for 3 correlators. It would also be nice to give the s parameter a good interpretation.

Finally, and arguably most importantly, it must be seen how (if) this result can be used to prove the main conjecture, which is an analogous statement about F . What makes the transition complicated is that there is a multiplication by a factor of $\exp((x^3 + y^3 + z^3)/24)$ whose series expansion relates terms whose exponents are off by multiples of 3. It is hoped that there is some geometric explanation for this exponential factor (as the $1/24$ is closely related to correlation numbers on genus 1 surfaces) which would allow for a proof without messy algebraic manipulation.

References

- [1] Zograf, P. (2020) *An Explicit Formula For Witten's 2-Correlators* <https://arxiv.org/abs/2012.03268v1>
- [2] Liu, K. & Xu, H. (2009) *Intersection Numbers and Automorphisms of Stable Curves*. Michigan Mathematical Journal
- [3] Liu, K. & Xu, H. (2009) *The N-Point Functions for Intersection Numbers on Moduli Spaces of Curves* <https://arxiv.org/abs/math/0701319>
- [4] Delecroix, V., Schmitt, J., & van Zelm, J. (2021) *admcycles - a Sage Package for Calculations in the Tautological Ring of the Moduli Space of Stable Curves*. Journal of Software for Algebra and Geometry